

Problem Set 4 (two pages)

Statistics 24510-30040 (W25)

Due 9 am Tuesday, February 4.

Reminder: In-class midterm on Thursday, February 6. (Details will be posted in Ed.)

Requirements

- Upload your pdf named "Lastname-Firstname-hw4.pdf" to Gradescope under Pset4. Be sure to mark the pages corresponding to each problem.
- Show your work. Provide detailed derivations and arguments in order to receive credit.
- You may discuss methods with others, but you should devise and write your answers by yourself completely and independently.

Problem assignments (Reference: Rice chapters 9, 11 and 12; Casella and Berger chapters 8 and 11.)

1. (*Two sample means: Pooled versus paired confidence intervals*)

Suppose $(X_1, Y_1), \dots, (X_n, Y_n)$ are i.i.d. bivariate normal, where n is not necessarily large. Assume $X_i \sim N(\mu_1, \sigma^2)$, $Y_i \sim N(\mu_2, \sigma^2)$. Note that only the means are different.

This problem compares two confidence intervals for $\mu_1 - \mu_2$, namely the *pooled interval* (ref. lectures or pp. 422-423 in Rice) and the *paired interval* (ref. lectures or pp. 446-447 in Rice).

- Find the lengths of the *paired* confidence interval and the *pooled* confidence interval.
Clearly define the terms used in your formula (e.g., variance estimator S_*^2 , t quantiles).
- Show that if the true correlation $\rho = \text{Corr}(X_i, Y_i) = 1$, then the expected length of the *paired interval* is zero.
(In fact the length is zero with probability one.)
- Assume now that the observations are such that the sample correlation coefficient is exactly equal to zero (so we assume $\rho = 0$).
Which confidence interval would have shorter expected length, the *paired interval* or the *pooled interval*?
Do the comparison by computing the ratio of the two expected square length of the intervals.

2. (*Binomial likelihood ratio test*)

Suppose random variables X_1 and X_2 are independent with $X_i \sim \text{Bin}(n_i, p_i)$. Assume the n_i 's are known.

We wish to test the hypothesis $H_o : p_1 = p_2$ versus the alternative hypothesis $H_a : p_1 \neq p_2$.

- Derive the likelihood ratio statistic LR for this hypothesis.
- What is the approximate distribution of the likelihood ratio test statistic $-2 \log(LR)$ under the null hypothesis assuming that n_1 and n_2 are both large? (Distribution parameters should be clearly specified.)
- Now consider applying this test to the ups and downs of the Dow Jones Industrial Average in the 20th century. During 1900-1949, there were 7810 up (including unchanged) days out of 14,736, and during 1950-1999, there were 6688 up days out of 12,673.
Under what assumptions on the sequence of up and down days would the null hypothesis in part (a) hold?
- Using the above and your result in part (a), find the numerical value for the likelihood ratio statistic for the hypothesis that the probability of an up day is equal for the two halves of the 20th century. Find the p-value for this test. What do you conclude?
(Note: you may use the R command `pchisq` to find the p-value.)

3. (Hands-on one-way ANOVA)

The leaves of a type of plants fold and unfold in various light conditions. A sample of 15 different leaves were treated with red light for 3 minutes. The leaves were randomly divided into three groups of five. The leaflet angles were then measured 30, 45 and 60 minutes after light exposure in the three groups.

The measurements are shown in the following table.

delay(minutes)	Angles (degrees)				
30	140	138	140	138	142
45	140	150	120	128	130
60	118	130	128	118	118

- What are the treatments? How many levels of treatment are there?
- Calculate the treatment sum of squares SS_T (a.k.a. SS_B) and the residual sum of squares SS_E (a.k.a. SS_W). Write out every term of the sum SS_T , and the first two and last two terms in SS_E .
- Create the ANOVA table for these data.
The columns of the table should include variance source, df, SS, MS, F-statistic, and p-value.
- Based on the above, test the null hypothesis that delay after exposure does not affect leaflet angle. (What is your alternative hypothesis?) interpret the result.

4. (MLE in one-way ANOVA under various constraints)

Consider the one-way ANOVA model parameterized as

$$Y_{ij} = \mu_i + \varepsilon_{ij}, \quad \varepsilon_{ij} \text{ i.i.d. } \sim N(0, \sigma^2), \quad j = 1, \dots, J_i, \quad i = 1, \dots, I.$$

Note that the sample sizes J_i 's are not necessarily equal.

- Write the likelihood function of the model (for observed data $y_{ij}, j = 1, \dots, J_i, i = 1, \dots, I$).
- Find the maximum likelihood estimates of the parameters $\mu_i \in \mathbb{R}, i = 1, \dots, I$.
- Find the maximum likelihood estimates of the parameter $\sigma^2 > 0$.
- Derive the likelihood ratio LR and state the LR test statistic for $H_o: \mu_1 = \dots = \mu_I$ (default H_a : not H_o).
- Compare the LR in (d) to the standard F test for one-way ANOVA. Find their relationship.
- Re-parameterize the model in the more common form

$$Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}, \quad \varepsilon_{ij} \text{ i.i.d. } \sim N(0, \sigma^2), \quad j = 1, \dots, J_i, \quad i = 1, \dots, I$$

with the constraint $\sum_{i=1}^I \alpha_i = 0$.

Find the maximum likelihood estimates of the parameters μ, α_i, σ^2 .

- Consider the same model as in (f),

$$Y_{ij} = \mu^* + \alpha_i^* + \varepsilon_{ij}, \quad \varepsilon_{ij} \text{ i.i.d. } \sim N(0, \sigma^2), \quad j = 1, \dots, J_i, \quad i = 1, \dots, I$$

now with the constraint $\alpha_1^* = 0$, as in R.

Find the MLE for μ^* and $\alpha_i^*, i = 2, \dots, I$, under the new constraint.